



INTEGRATEDCIRCUIT

TECHNICAL DATA

TA7136AP

TOSHIBA BIPOLAR LINEAR INTEGRATED CIRCUIT

SILICON MONOLITHIC

PRE-AMPLIFIER FOR DUAL POWER SUPPLY

VARIOUS HIGH QUALITY PRE-AMPLIFIER

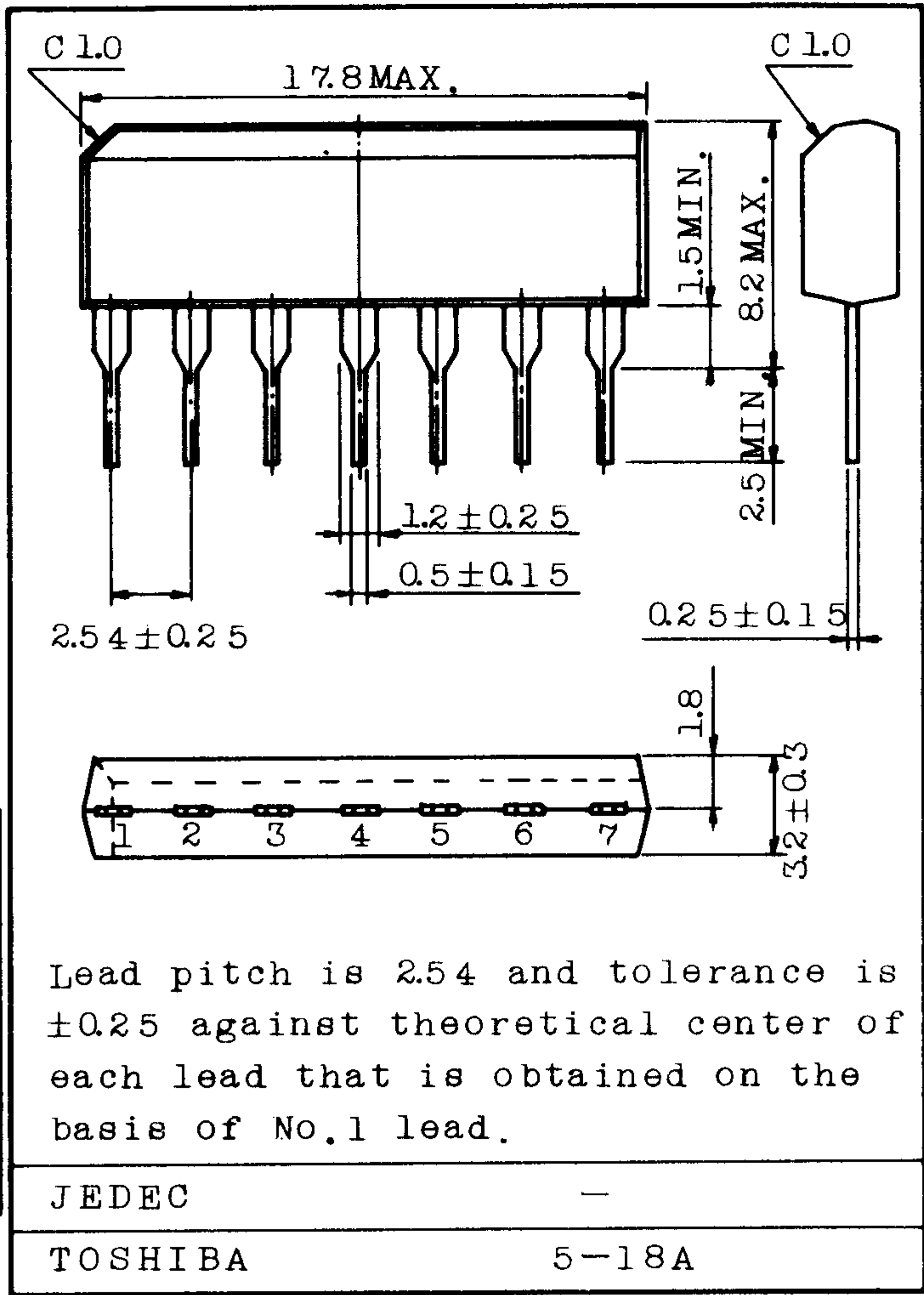
Unit in mm

- Low Noise : $V_{NI}=0.8\mu V_{rms}(Typ.)$
- High Open Loop Voltage Gain : $G_{VO}=92dB(Typ.)$
- Low Distortion : $THD=0.1\%(Max.)$
(RIAA. EQ. 40dB(1kHz), $V_{OUT}=7V_{rms}$)
- Wide Operating Supply Voltage Range : $V_{CC}=\pm 3\sim\pm 20V$

MAXIMUM RATINGS (Ta=25°C)

CHARACTERISTIC	SYMBOL	RATING	UNIT
Supply Voltage	V7-V4	40	V
Power Dissipation (Note)	P _D	400	mW
Operating Temperature	T _{opr}	-25 ~ 75	°C
Storage Temperature	T _{stg}	-55 ~ 125	°C

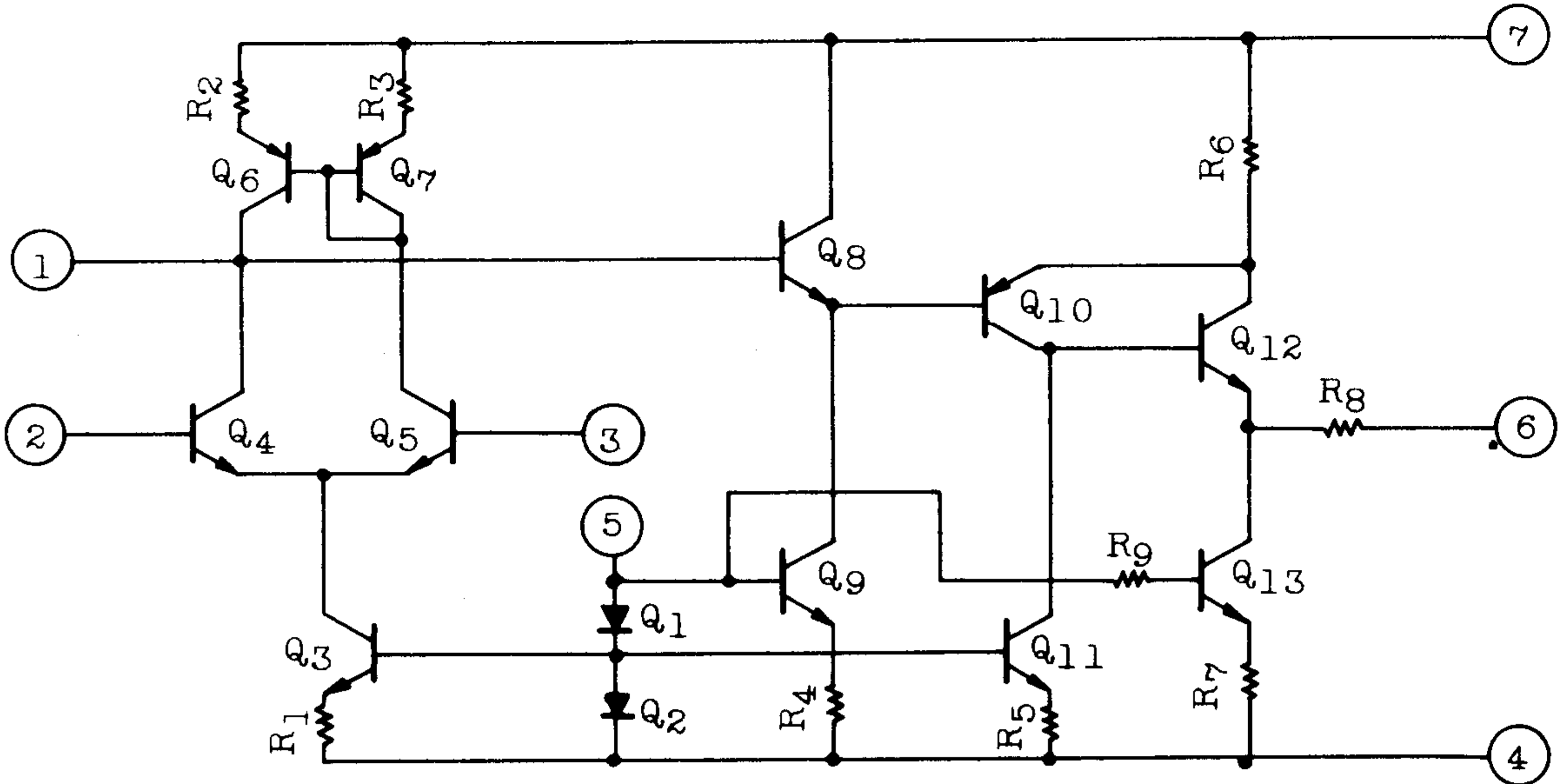
Note : Derated above Ta=25°C in the proportion of 4 mW/°C.



ELECTRICAL CHARACTERISTICS (V_{CC}=15V, V_{EE}=-15V, Ta=25°C)

CHARACTERISTIC	SYMBOL	TEST CIR-CUIT	TEST CONDITION	MIN.	TYP.	MAX.	UNIT
Supply Current	I _{CC}	1	V _{IN} =0	-	3.1	4.2	mA
Voltage Gain (Open Loop)	G _{VO}	1	f=1kHz, V _{IN} =-85dBm	87	92	-	dB
Maximum Output Voltage	V _{OM}	2	RIAA EQ, f=1kHz, THD=0.1%	7.0	-	-	V _{rms}
Equivalent Input Noise Voltage	V _{NI}	3	RIAA equalizer R _g =2.2kΩ, f=1kHz	-	0.8	1.5	μV _{rms}

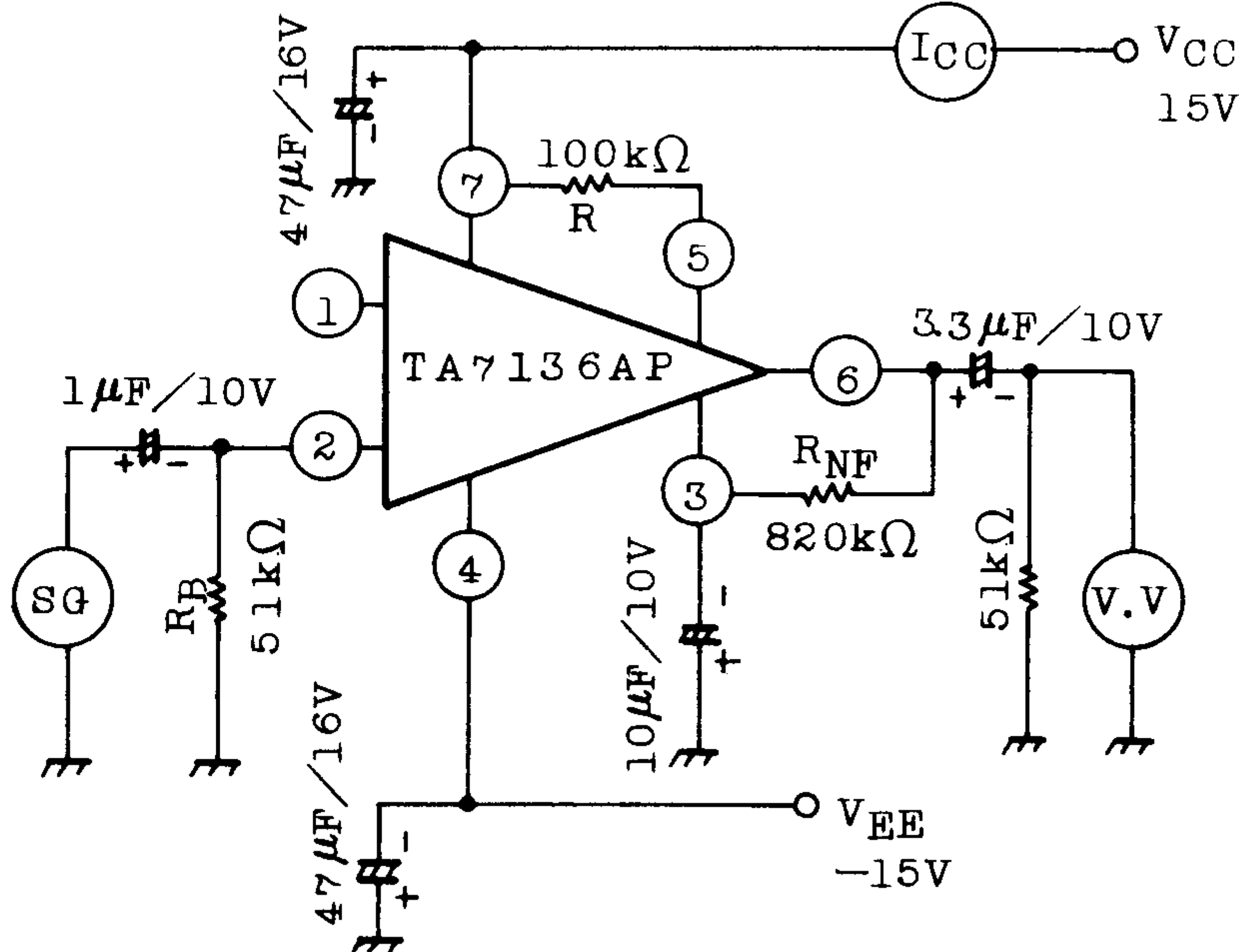
EQUIVALENT CIRCUIT



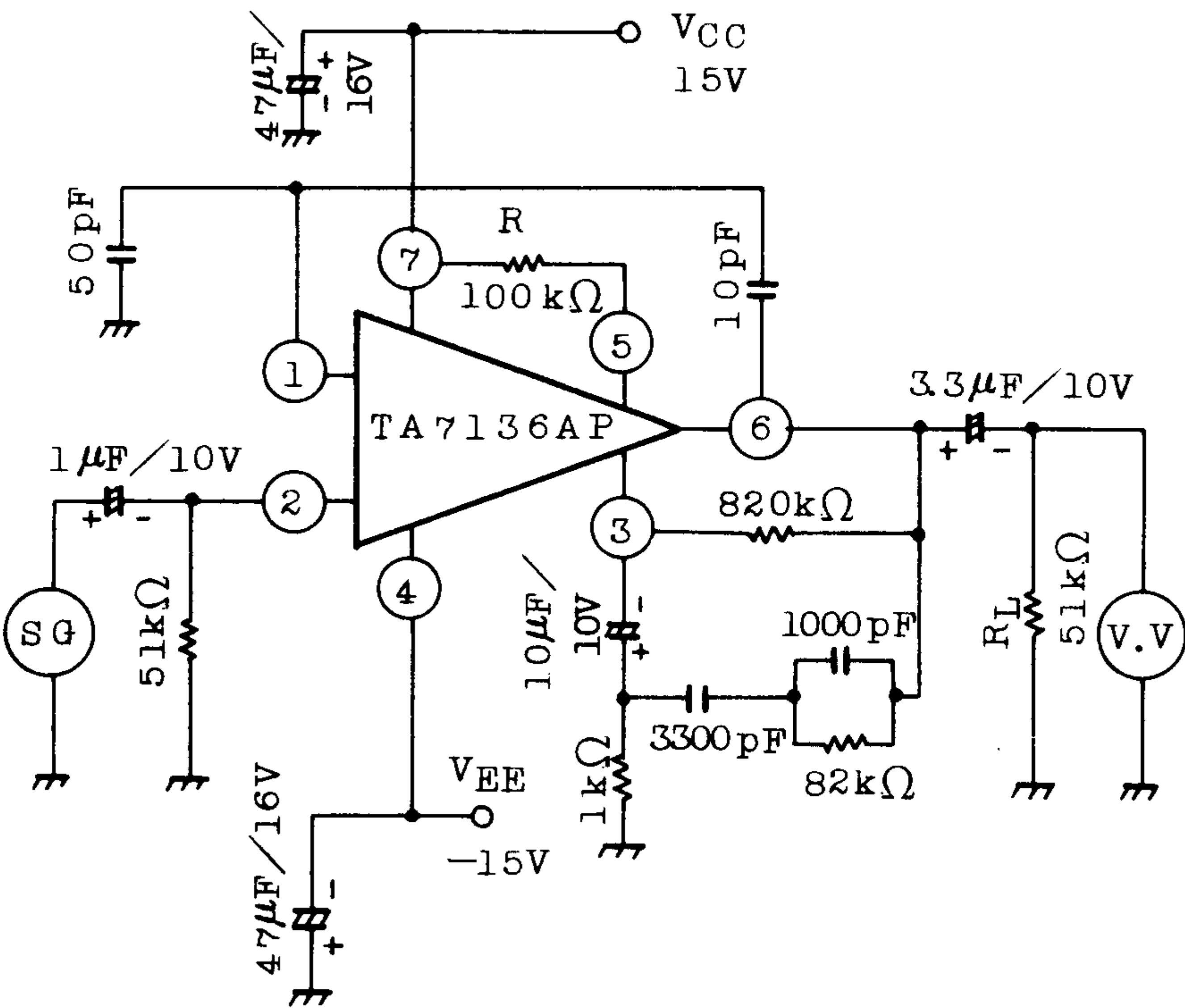
TECHNICAL DATA

TEST CIRCUIT

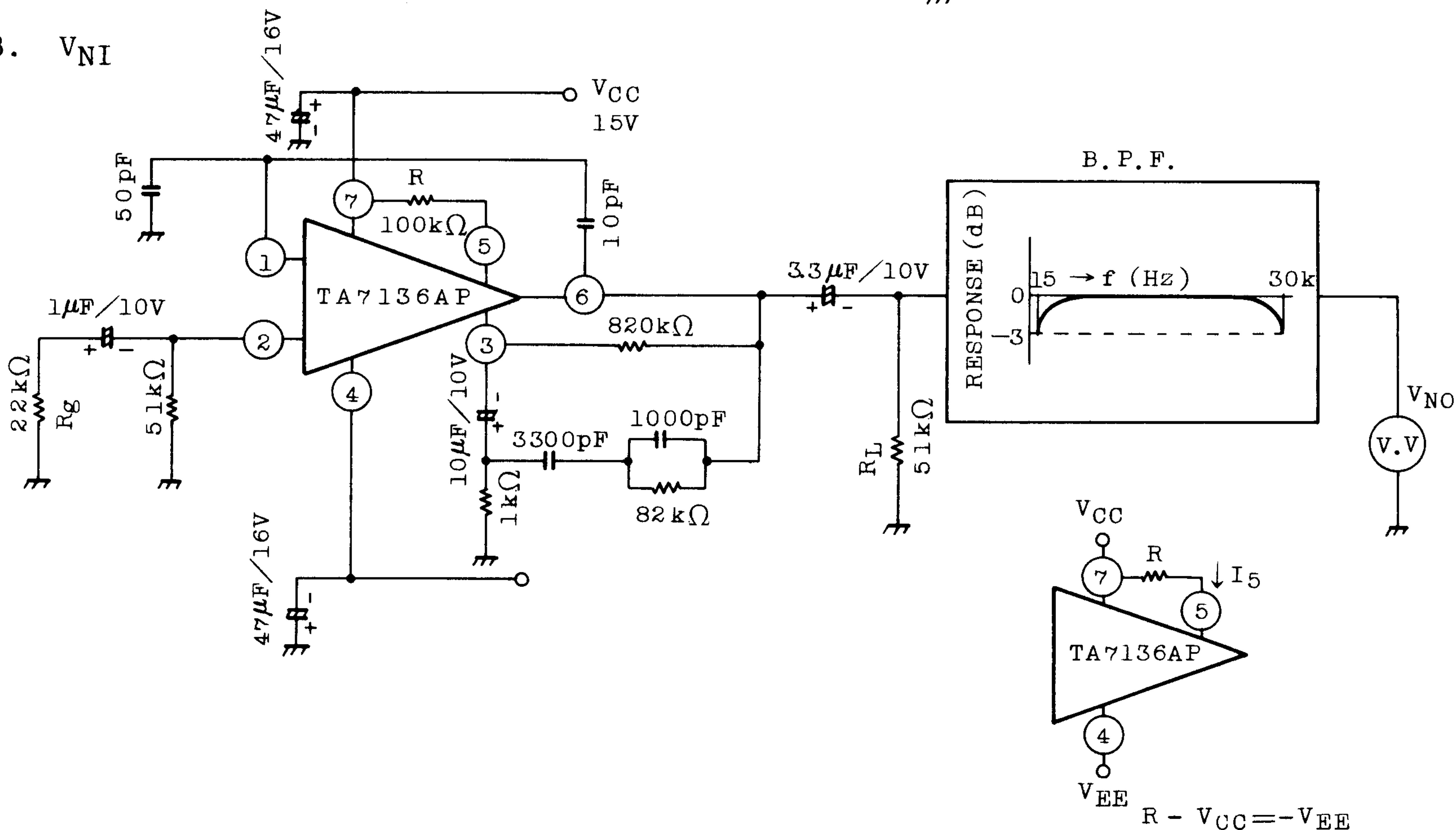
1. I_{CC} , G_{VO}



2. V_{OM}



3. V_{NI}



DECISION OF BIAS RESISTANCE R

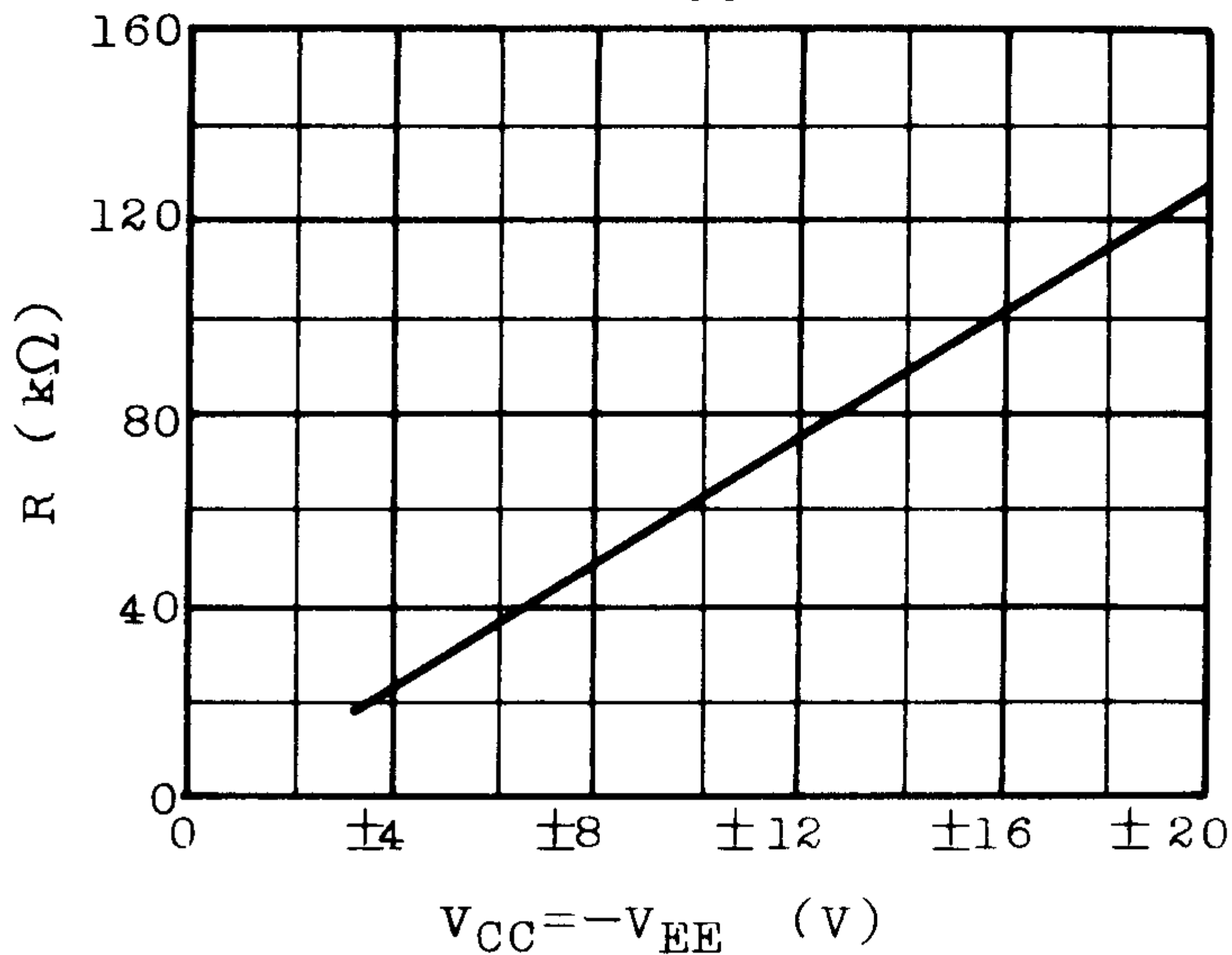
The TA7136AP is designed to operate under the bias condition $I_5 = 300\mu A \pm 20\%$.

Please decide the R by the following equation.

$$I_5 = (V_{CC} - V_{EE} - 2V_F) / R$$

$$R = (V_{CC} - V_{EE} - 1.4) / 0.3 (k\Omega)$$

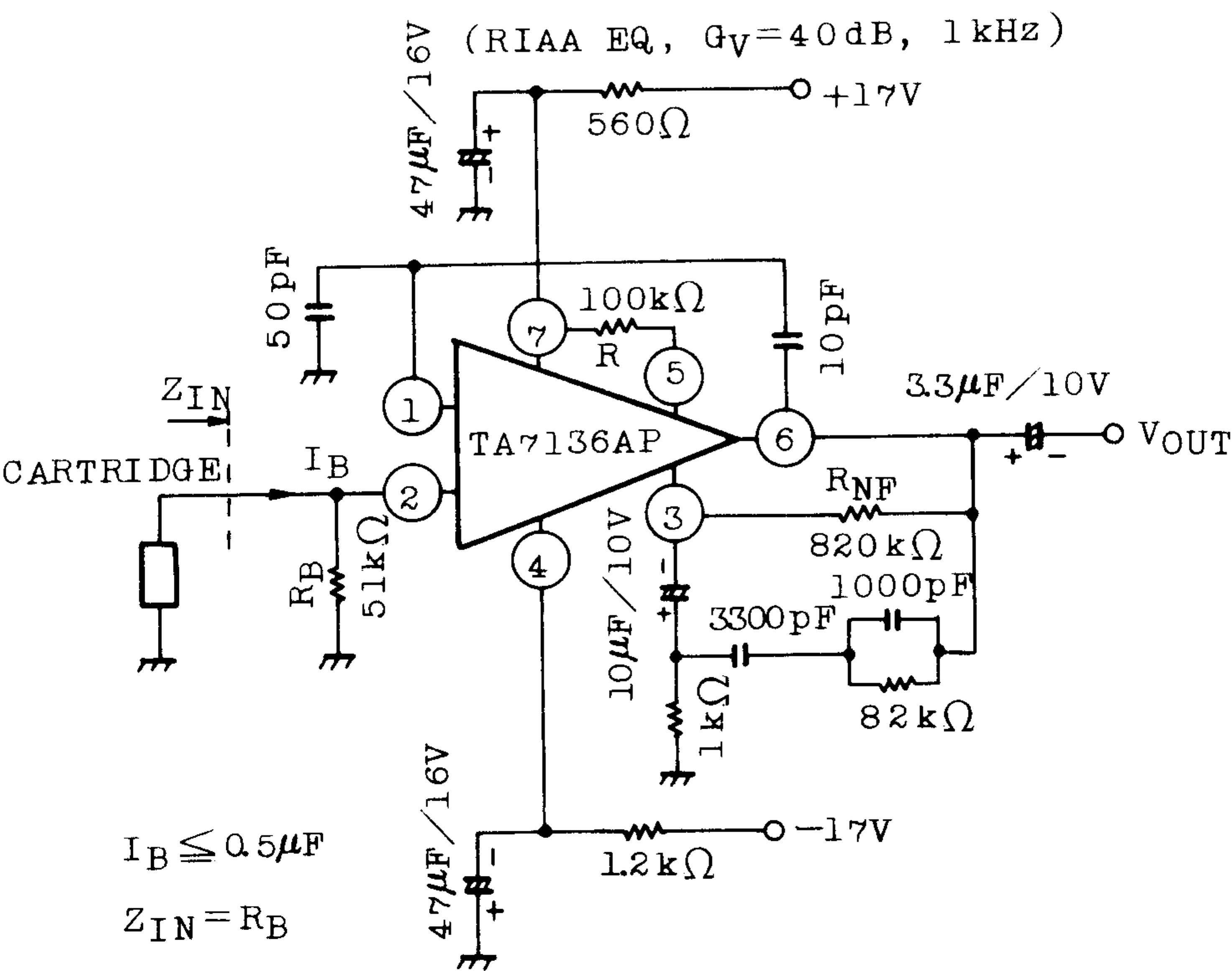
The following figure shows the calculated value of R.



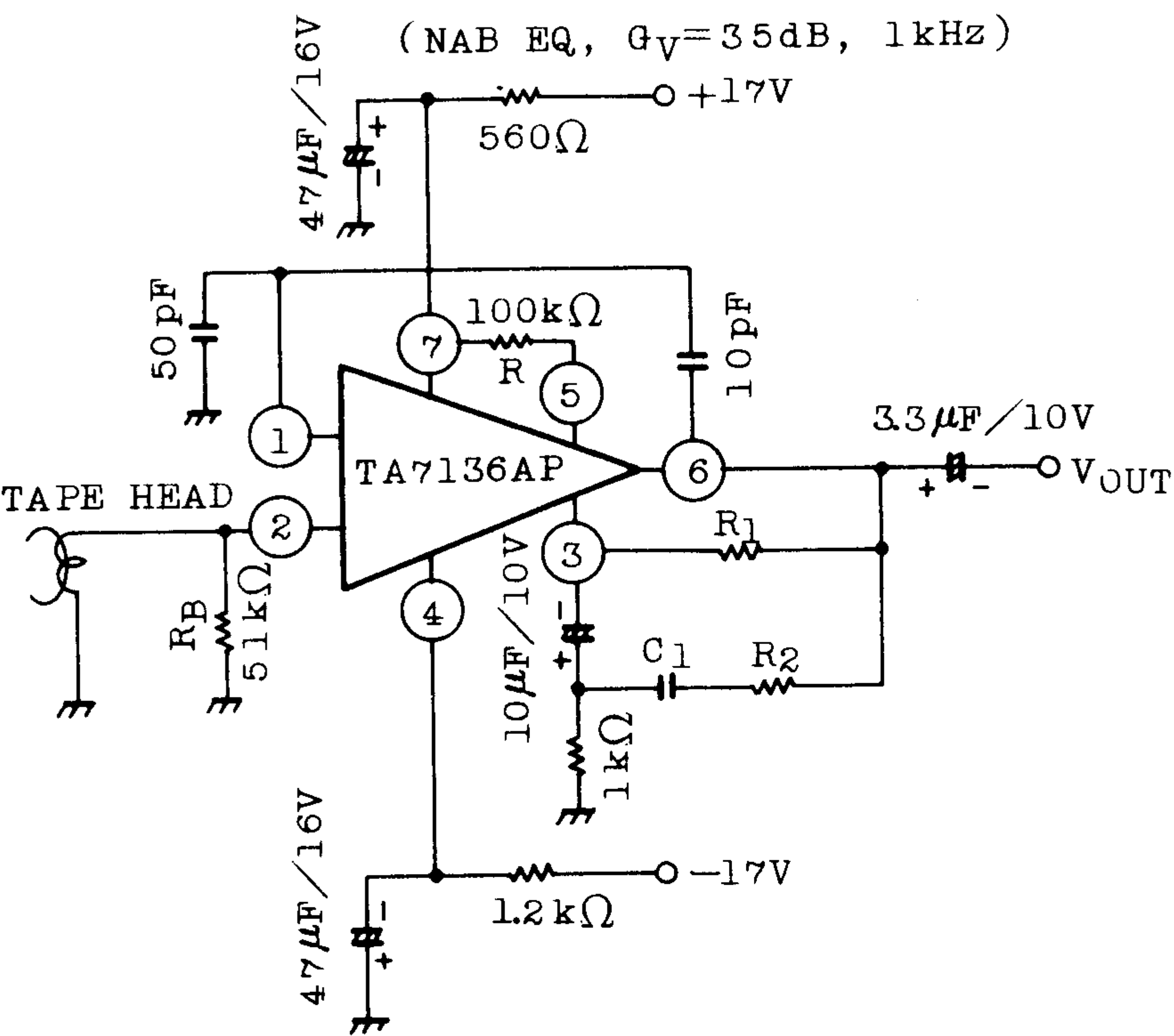


APPLICATION CIRCUIT

1. MAGNETIC PHONO PRE-AMPLIFIER

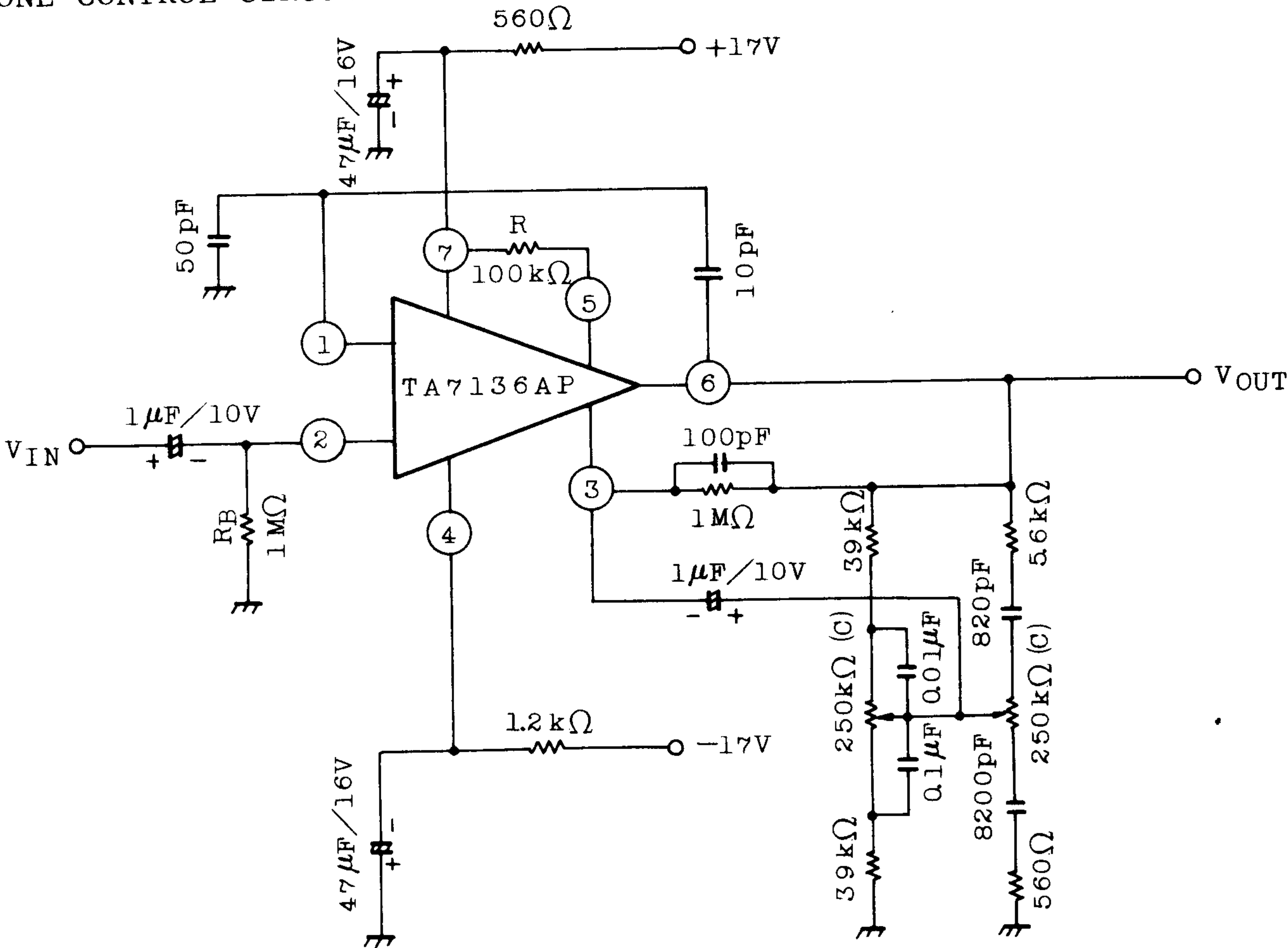


2. TAPERECORDER PRI-AMPLIFIER



	9.5cm/sec	19cm/sec	CASSETTE
R1	910kΩ	1MΩ	510kΩ
R2	27kΩ	18kΩ	47kΩ
C1	3300pF	2800pF	3300pF

3. TONE CONTROL CIRCUIT

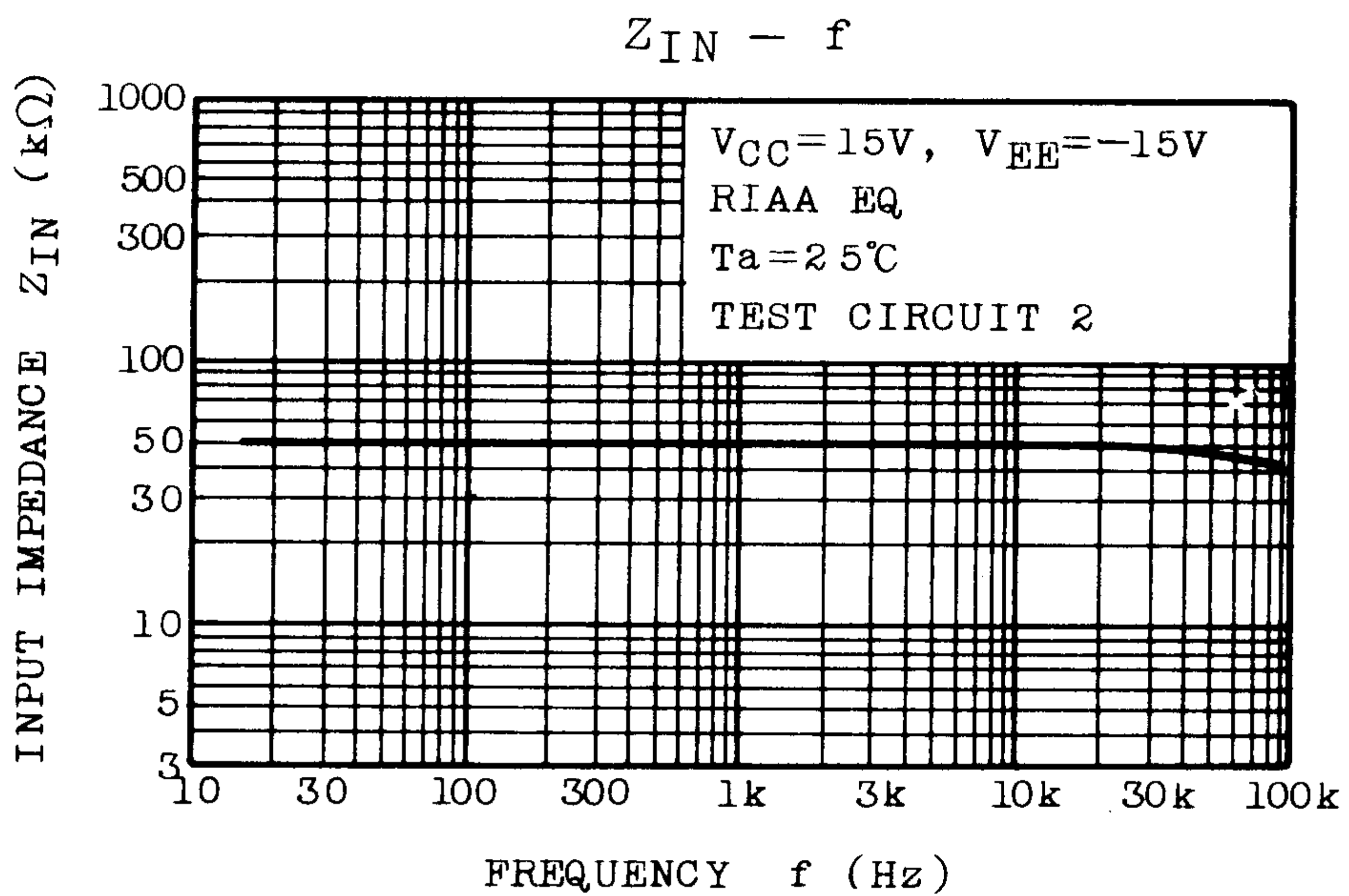
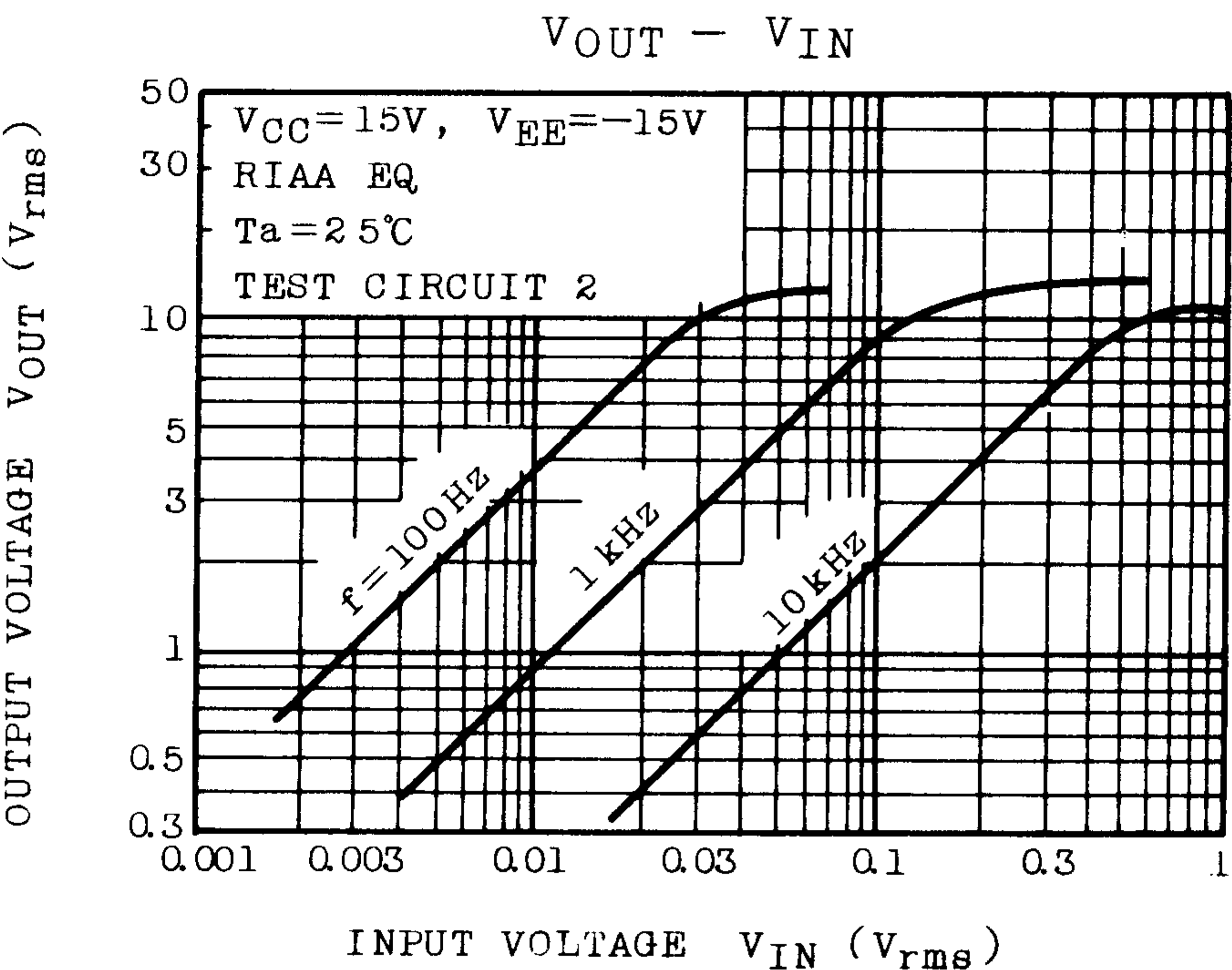
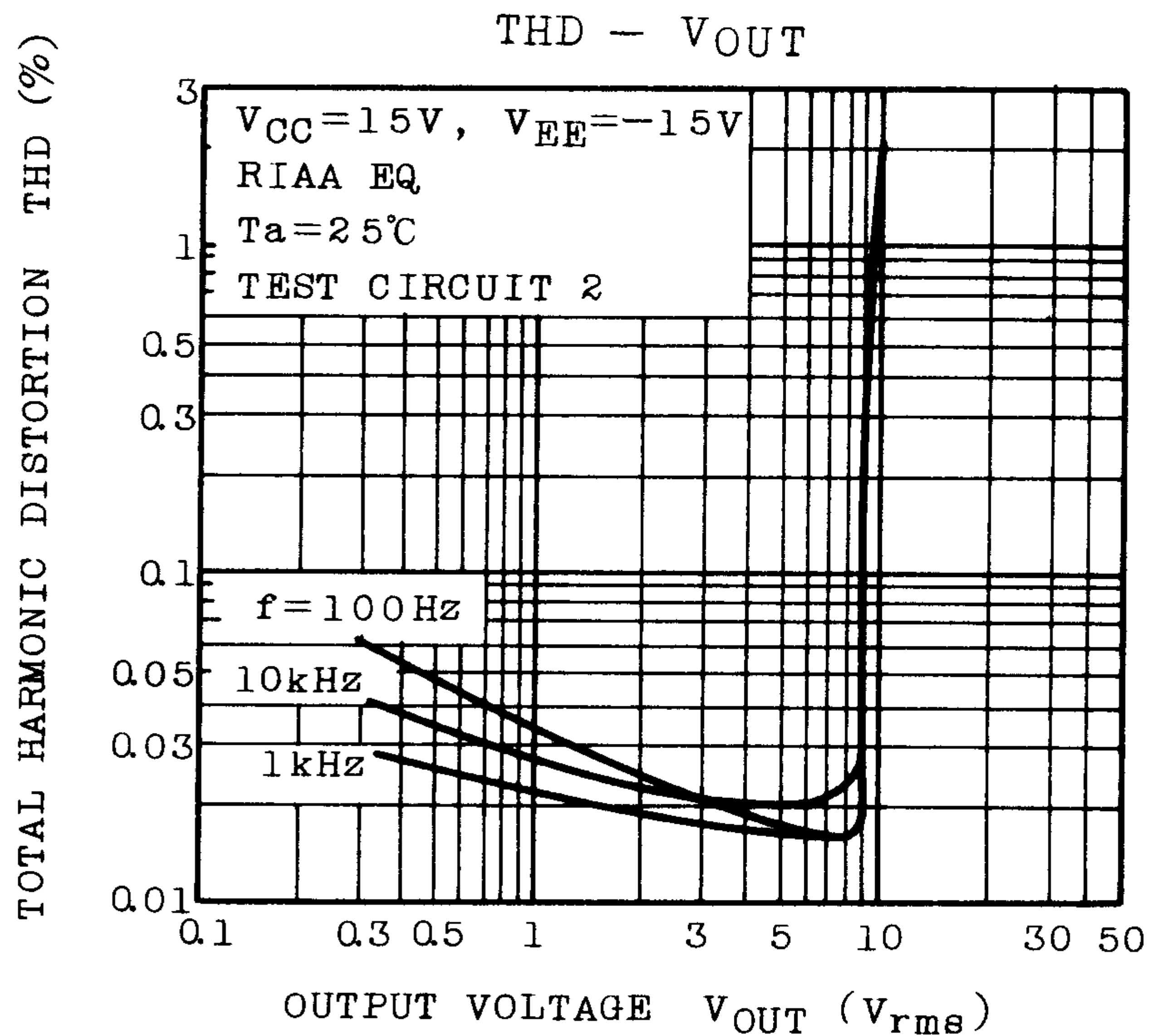
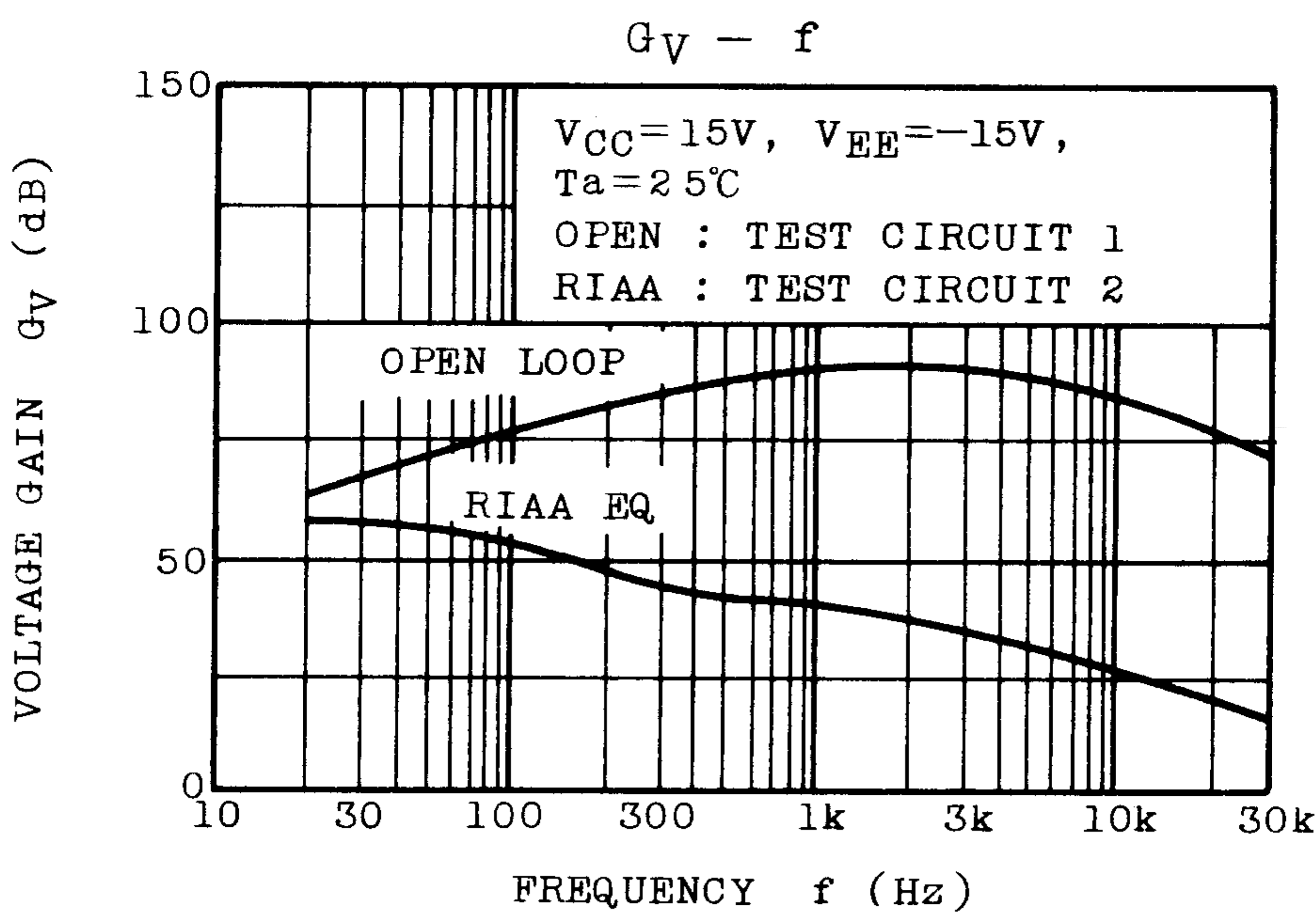
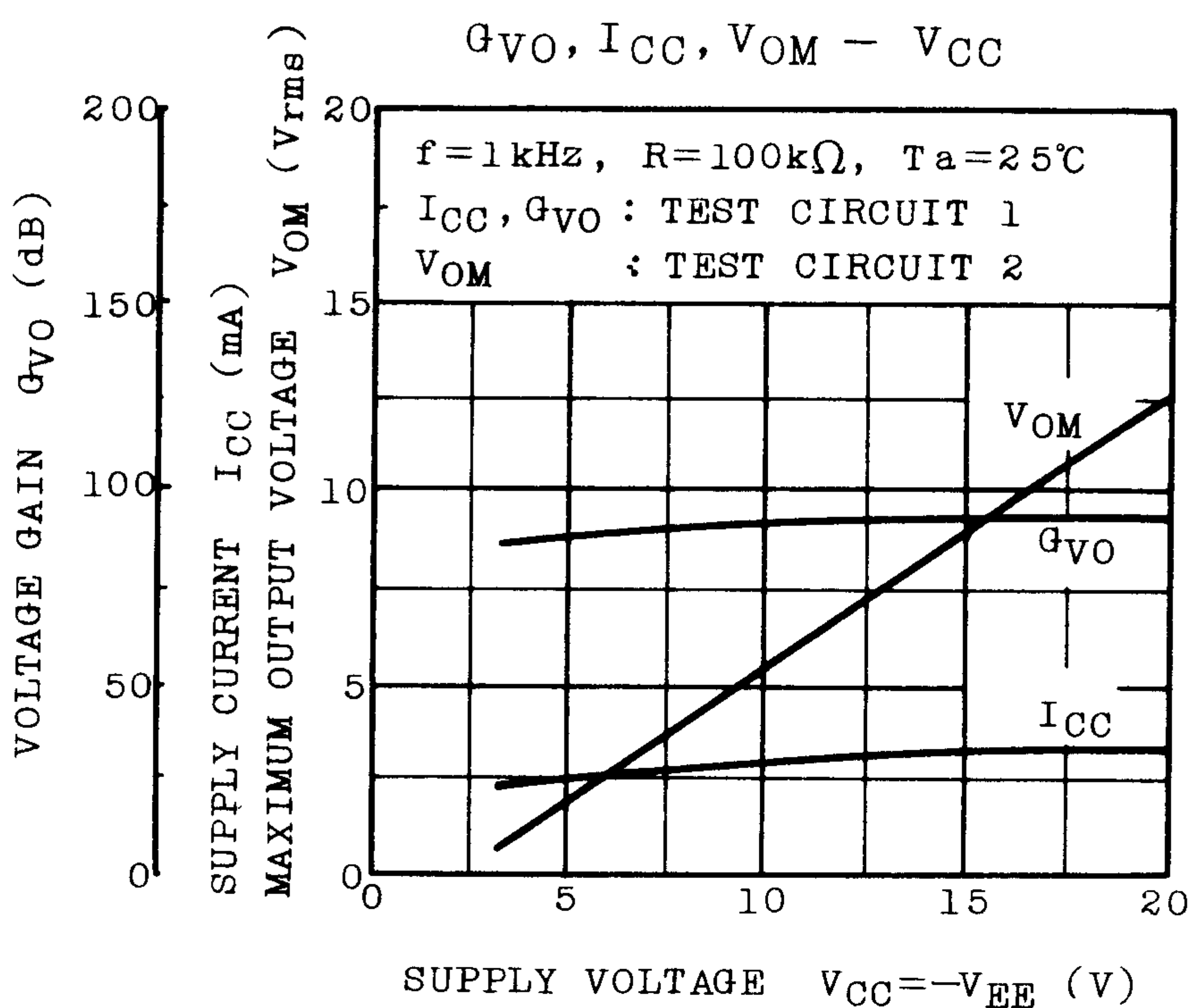
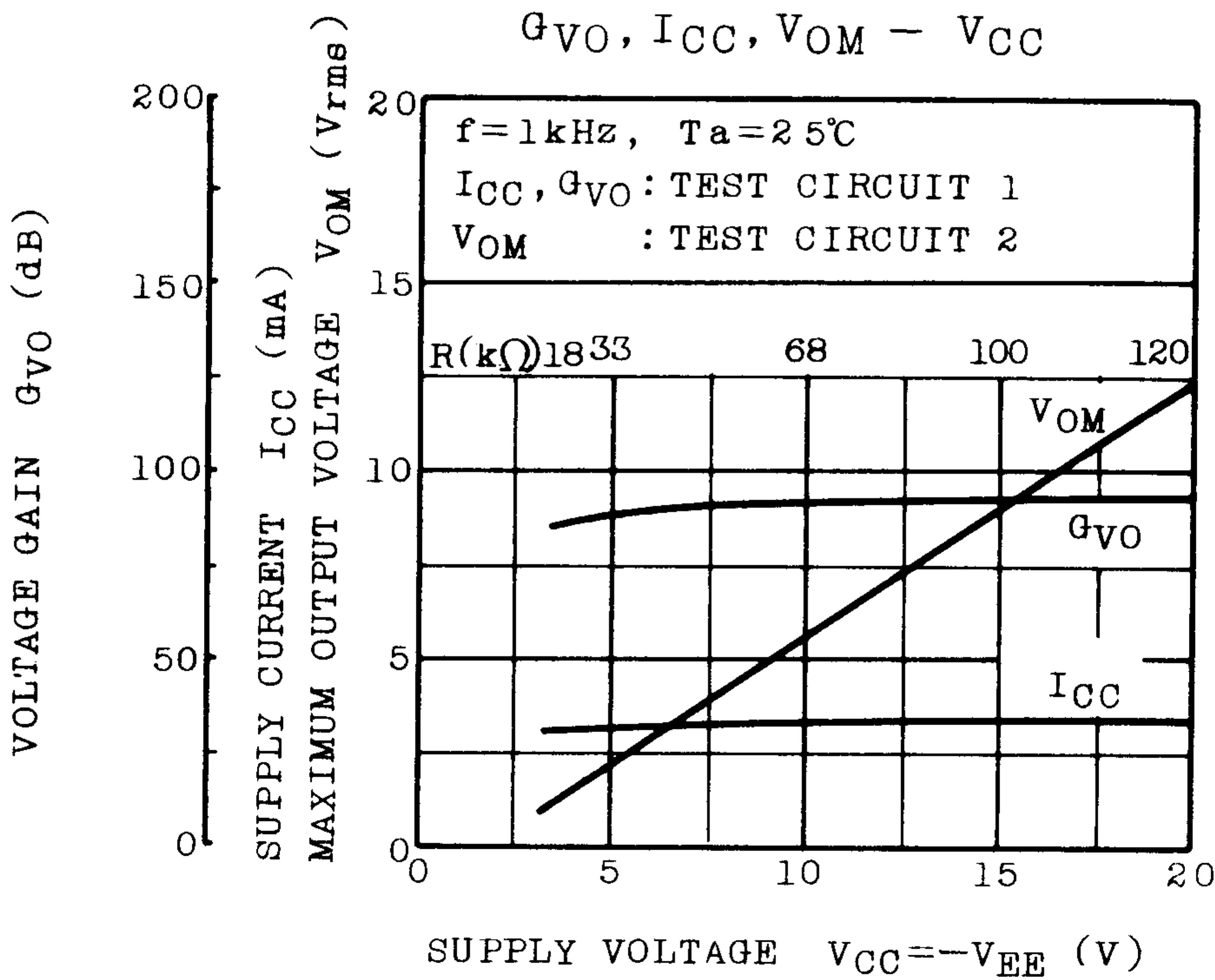




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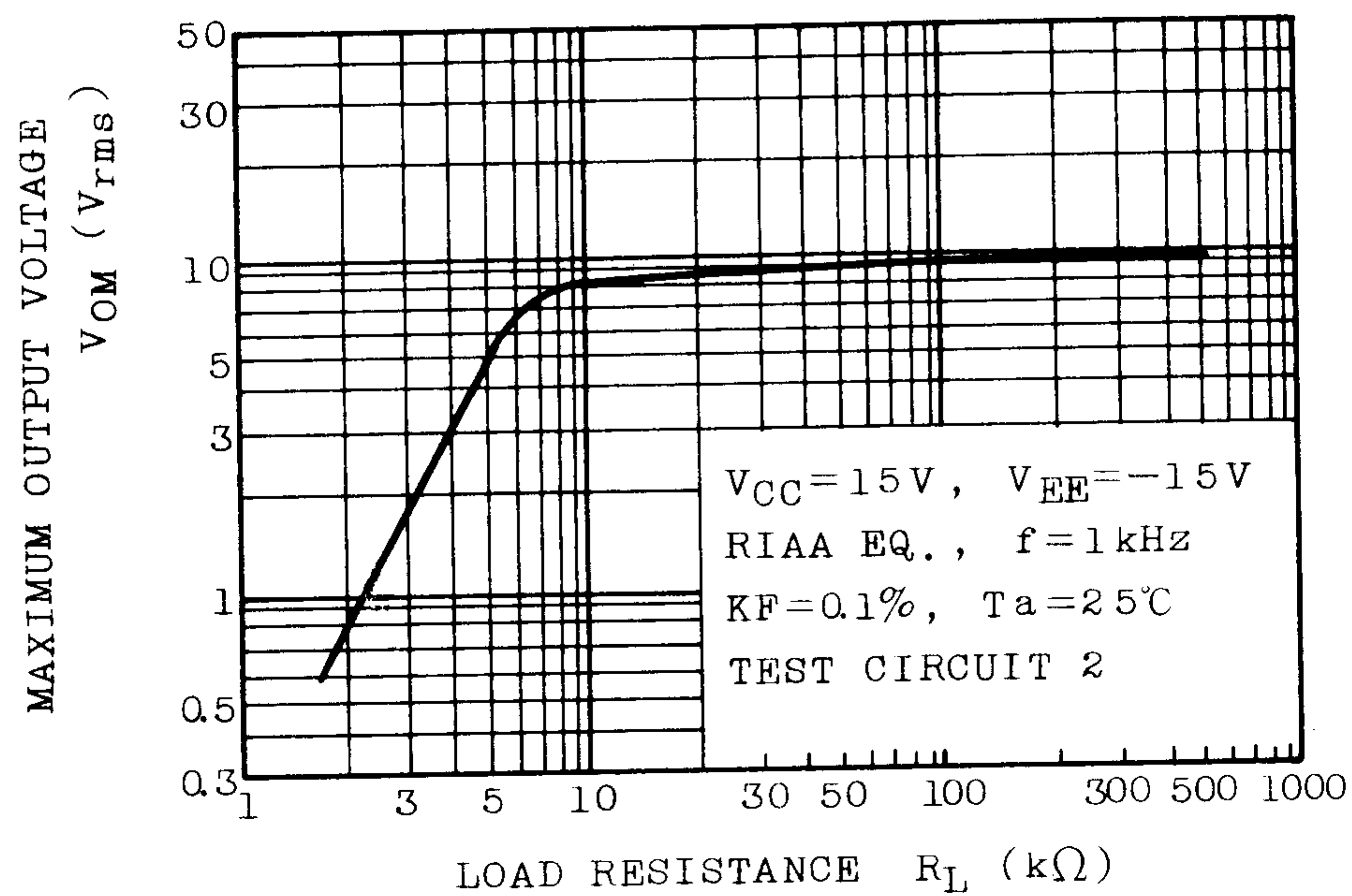
TA7136AP



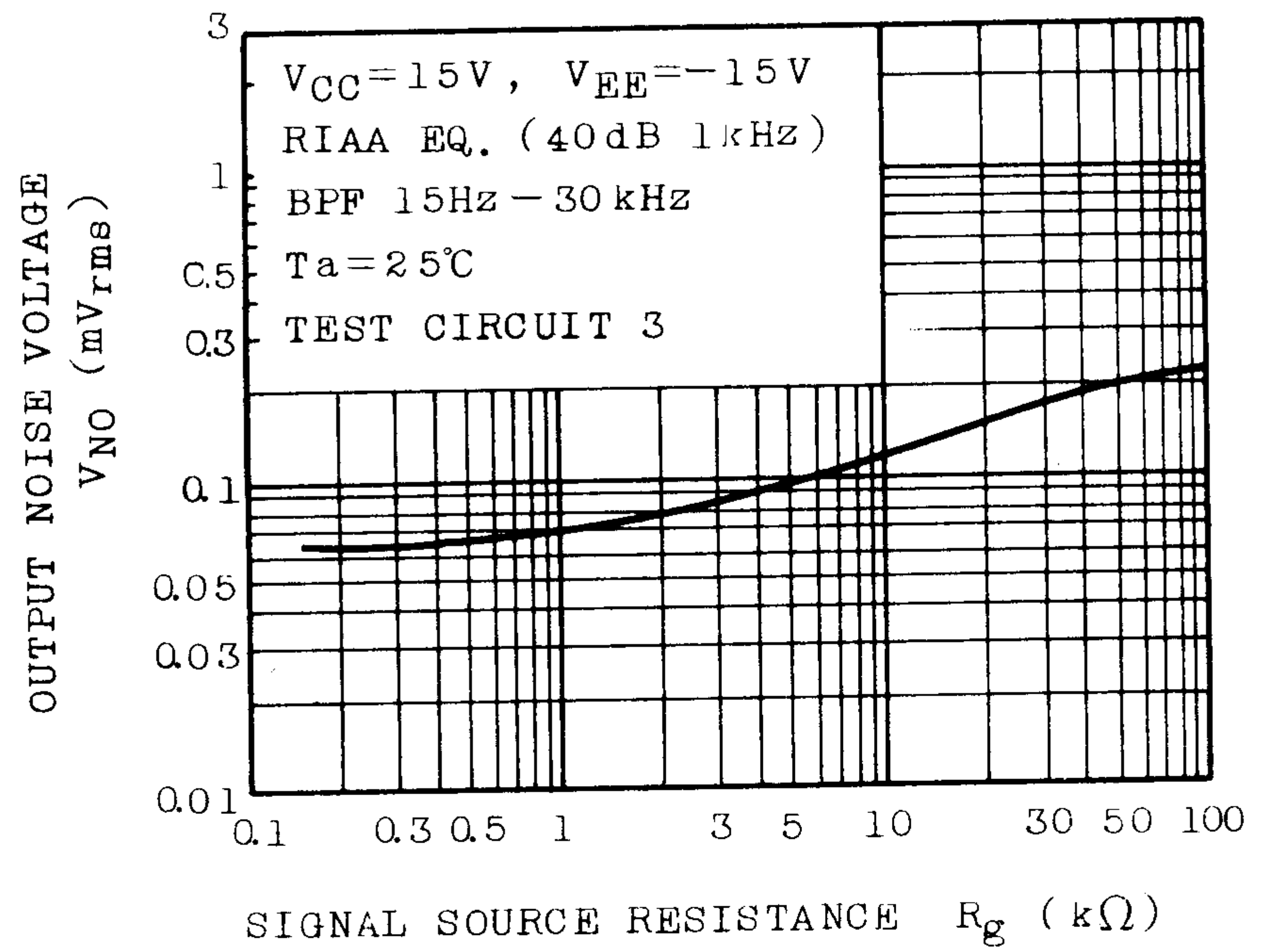


TECHNICAL DATA

$V_{OM} - R_L$



$V_{NO} - R_g$



$G_{VO}, I_{CC}, V_{OM} - T_a$

